

SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade
Kuniamuthur, Coimbatore - 641008

Possession of any form of Electronic gadgets even in OFF mode is treated as Malpractice

END SEMESTER EXAMINATIONS – Nov/Dec 2025

Regular

Regulations – 2025

PROGRAMME(S)	SEMESTER	COURSE CODE	COURSE TITLE
B.E/B.Tech: AI&ML, CIVIL, CSBS, CYBER, ECE, EEE, IOT, M.TECH - CSE, MECH, MT	1	25MSC01	Materials Science

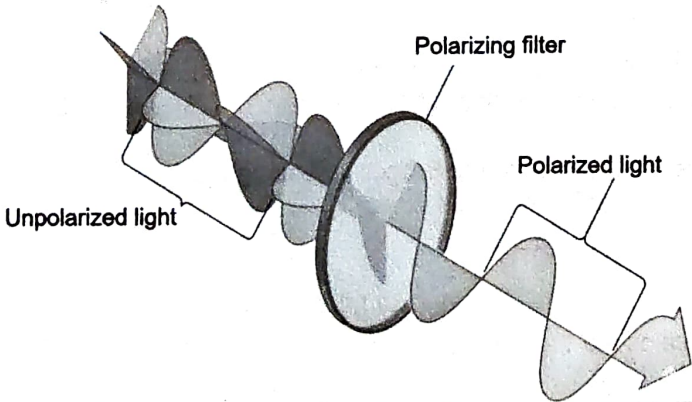
Maximum Marks : 100 Marks

Duration : Three Hours

COURSE OUTCOMES			BT
CO1	To apply the principles of wave optics in practical and theoretical contexts.		AP
CO2	Apply quantum principles to explain atomic and subatomic system behavior.		AP
CO3	Understand the principles of Lasers, Einstein's coefficients and the fundamentals of fiber optics along with their practical applications.		U
CO4	Apply the principles of semiconducting materials to analyze and design electronic circuits and systems.		U
CO5	Utilize the fundamental principles of nanomaterials to explore and implement their applications in Engineering.		AP

ANSWER ALL QUESTIONS

PART A (10 x 2 Marks = 20 Marks)

		BT	CO	MARKS
1	Illustrate the property of light taking place in the depicted figure. 	AP	CO1	2
2	Two narrow slits 0.08 cm apart are illuminated by light of frequency 8×10^{11} kHz. It is desired to have a fringe width of 6×10^{-4} m. Where should the screen be placed from the slit?	AP	CO1	2
3	Given a wave function ψ , how will you check whether it is well behaved wave function?	AP	CO2	2
4	An electron is trapped in a one dimensional box of width 0.1 nm. Calculate the energy of the electron in its ground state.	AP	CO2	2

5	Calculate the wavelength of radiation emitted by an LED made up of a semiconducting material with bandgap of 2.8 eV.	U	CO3	2
6	Differentiate step index and graded index fibres.	U	CO3	2
7	State the law of mass action in a semiconductor and write its importance.	U	CO4	2
8	Differentiate diffusion current from drift current.	U	CO4	2
9	What happens when a bulk material is transformed into a nano material?	AP	CO5	2
10	What are all the methods to prepare carbon nanotubes and why CNT are more strong?	AP	CO5	2

ANSWER ALL QUESTIONS

PART B (5 x 16 Marks = 80 Marks)

			BT	CO	MARKS
11	i)	You are given a transparent thin film of some thickness in the range of few thousand angstrom units. Explain the method to find the thickness of the thin film using an interferometer.	AP	CO1	8
	ii)	Compare Fresnel's diffraction and Fraunhofer's diffraction. Light of wavelength 500 nm passes through a slit of 0.2 mm wide. The diffraction pattern is formed on a screen 60 cm away. Determine the angular spread of central maximum.	AP	CO1	8

OR

12	i)	Explain how diffraction happens through a single slit through a neat diagram.	AP	CO1	8
	ii)	Write the difference between interference and diffraction. And find the width of the slit, if a parallel beam of light of wavelength 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 cm away. it is observed that the first minimum is at a distance of 2.5 mm from the centre of the screen.	AP	CO1	8
13	i)	Derive Schrodinger's time independent and time dependent wave equation. And also calculate the wavelength associated with neutron of energy 0.025 eV	AP	CO2	10
	ii)	List the properties of matter waves and derive the deBroglie wavelength of an electron being accelerated by a potential of V volts	AP	CO2	6

OR

14	i)	Derive an expression for energy levels and wave function of a particle in one dimensional potential well. And calculate the lowest energy of an electron confined in a potential well of width 1.5 Å	AP	CO2	10
	ii)	State Heisenberg's uncertainty principle. And using this principle prove that electrons cannot reside inside the nucleus.	AP	CO2	6
15	i)	Derive Einstein's relation between the stimulated emission and spontaneous emission.	U	CO3	8
	ii)	Explain the working principle behind the CO ₂ laser. Draw a neat diagram of its vibrational modes and energy levels.	U	CO3	8

OR

16	i)	Explain how light enters and propagates in an optical fiber.	U	CO3	8
	ii)	How fiber optics are classified based on refractive index profile and modes of propagation.	U	CO3	8

17	i)	Explain how bands are formed in a solid and draw the energy band structure of a P and N type semiconductor.	U	CO4	8
	ii)	How a PN- junction is formed, explain the depletion region, potential barrier and diffusion current with schematic and energy level diagram.	U	CO4	8

OR

18	i)	Explain CE, CB and CC mode in a Bipolar Junction Transistor.	U	CO4	8
	ii)	Select one mode of BJT with high gain and explain the transistor action in it.	U	CO4	8

19	i)	List any 4 properties and 4 applications of nanomaterials.	AP	CO5	8
	ii)	Explain the preparation of nanomaterials by sol-gel technique and what are the advantage of this method?	AP	CO5	8

OR

20	i)	Select one method for the preparation of Carbon Nano Tubes and explain with a neat diagram.	AP	CO5	8
	ii)	What do you mean by electrochemical deposition? What type of nanomaterials can be prepared by this method, explain the technique?	AP	CO5	8

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